

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA0604	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	G. Hari Prasad	Reg. No.:	192525273
Section / Batch:	B	Date:	05/08/2026

Question Type	Focus Area	Questions	Marks
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1 – Q2	100 (2 × 50)
GRAND TOTAL:			100 Marks

SECTION A – SCENARIO BASED ASSESSMENT

Instructions to Students

- Answer BOTH scenario-based questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Code of Conduct

I **G. Hari Prasad (192525273)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *G. Hari Prasad*

Q36.

A retail chain maintains a product list that is frequently updated and unsorted. The system searches for products using Linear Search. a) Explain how Linear Search operates in this system. b) Analyze its performance as dataset size increases. c) Suggest whether Binary Search would be beneficial and justify.

Program Python Code:

```
def linear_search(products, target_id):
    for i in range(len(products)):
        if products[i]['id'] == target_id:
            return i
    return -1

products = [
    {'id': 101, 'name': 'Laptop'},
    {'id': 205, 'name': 'Headphones'},
    {'id': 310, 'name': 'Keyboard'},
    {'id': 415, 'name': 'Monitor'},
    {'id': 520, 'name': 'Mouse'}
]

target_id = 310
index = linear_search(products, target_id)

if index != -1:
    print(f"Product Found: {products[index]['name']} at index {index}")
else:
    print("Product not found")
```

Output:

```
PS C:\Users\hariprasad\Desktop\DAA> python q36.py
Product Found: Keyboard at index 2
PS C:\Users\hariprasad\Desktop\DAA>
```

Q76.

An online retail system searches for customer orders stored in an unsorted list using Linear Search. a) Explain how Linear Search is applied. b) Analyze performance degradation as data grows. c) Suggest improvements and justify.

Program Python Code:

```
def linear_search_order(orders, target_order_id):
    for i in range(len(orders)):
        if orders[i]['order_id'] == target_order_id:
            return i
    return -1

orders = [
    {'order_id': 'ORD1001', 'customer': 'Ravi'},
    {'order_id': 'ORD1002', 'customer': 'Meena'},
    {'order_id': 'ORD1003', 'customer': 'Arjun'},
    {'order_id': 'ORD1004', 'customer': 'Divya'},
    {'order_id': 'ORD1005', 'customer': 'Kiran'}
]

target_order_id = 'ORD1004'
index = linear_search_order(orders, target_order_id)

if index != -1:
    print(f"Order Found: {orders[index]['customer']} at index {index}")
else:
    print("Order not found")
```

Output:

```
PS C:\Users\hariprasad\Desktop\DAA> python q76.py
Order Found: Divya at index 3
PS C:\Users\hariprasad\Desktop\DAA>
```